INDUCED LORENTZIAN POLYNOMIALS

CHRISTOPHER EUR, NUTAN NEPAL, DANIEL QIN

ABSTRACT. Given a party of m people, whose expertise collectively covers n topics, and a subset T of the topics, consider forming a panel of $|T|$ people from the party such that T can be covered by assigning a distinct topic to each panel member with the expertise. We show that the numbers of such panels, as T varies, form a Lorentzian polynomial. We achieve this by showing that a certain linear operator on polynomials, which we call the "inducing operator" for its connection to induced (poly)matroids, preserves Lorentzian polynomials and realizable volume polynomials.

1. INTRODUCTION

For a positive integer m, let $[m] = \{1, \dots, m\}$, and let $\mathcal{S} = (S_1, \dots, S_n)$ be a finite sequence of subsets of $[m]$. For a subset $T \subseteq [n]$, consider the following matching statistic:

$$C_T(\mathcal{S}) := |\{B \subseteq [m] : \text{there is a bijection } m : T \to B \text{ with } m(t) \in S_t \text{ for all } t \in T\}|.$$

We show that these quantities display the following log-concavity property. For a nonnegative integer r, let $\binom{[n]}{r}$ be the set of cardinality-r subsets of $[n]$, and for $T \subseteq [n]$, write $y^T = \prod_{t \in T} y_t$.

Theorem 1.1. For any integer $0 \leq r \leq m$, the polynomial $f_{\mathcal{S},r} := \sum_{T \in \binom{[n]}{r}} C_T(\mathcal{S}) y^T$ is *Lorentzian* in the sense of **[?]**. More strongly, it is a *realizable volume polynomial* over any field in the sense of **[?]**.

See **[?]** for a survey of Lorentzian polynomials and volume polynomials, as well as inequalities that their coefficients satisfy. In particular, the theorem implies that the quantities $C_T(\mathcal{S})$ satisfy variations of the Alexandrov–Fenchel inequality and the reverse Khovanskii–Teissier inequality. We do not know elementary proofs of such inequalities for the $C_T(\mathcal{S})$.

We caution that a subset B is counted towards $C_T(\mathcal{S})$ only once even if the number of bijections m satisfying the condition is greater than one. As a result, a partial derivative of $f_{\mathcal{S},r}$ is usually not another polynomial constructed with a similar matching statistic. The common inductive strategy involving partial derivatives for showing that a polynomial is Lorentzian therefore does not seem useful in our case. We deduce the theorem by introducing a linear operator on polynomials that preserve Lorentzian polynomials and realizable volume polynomials, as follows. Let $\Gamma_{\mathcal{S}}$ be a bipartite graph on vertices $[m] \sqcup [n]$ where a vertex $i \in [m]$ is connected to a vertex $j \in [n]$ if and only if $i \in S_j$.

Definition 1.2. We say a pair of $\alpha \in \mathbb{Z}^m_{\geq 0}$ and $\beta \in \mathbb{Z}^n_{\geq 0}$ admits a *$\mathcal{S}$-matching* if there exists a weighting $w : E(\Gamma_{\mathcal{S}}) \to \mathbb{Z}_{\geq 0}$ of the edges of $\Gamma_{\mathcal{S}}$ such that, for all $i \in [m]$ and $j \in [n]$, the sum of the weights of the edges incident to i is equal to α_i, and to j is equal to β_j. We write $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$ in this case.

For $\alpha \in \mathbb{Z}^m_{\geq 0}$, write $\alpha! = \alpha_1! \cdots \alpha_m!$ and $x^\alpha = x_1^{\alpha_1} \cdots x_m^{\alpha_m}$.

Theorem 1.3. The linear operator $\mathcal{I}_{\mathcal{S}} : \mathbb{R}[x_1, \ldots, x_m] \to \mathbb{R}[y_1, \ldots, y_n]$ defined on the monomials by

$$\mathcal{I}_{\mathcal{S}} \left(\frac{x^\alpha}{\alpha!} \right) := \sum_{\beta : \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta} \frac{y^\beta}{\beta!}$$

preserves Lorentzian polynomials and realizable volume polynomials over a field.

The operator $\mathcal{I}_{\mathcal{S}}$ does not always preserve *stable polynomials* in the sense of [**?**]; see Example 2.5. We prove Theorem 1.3 in Section 2 by showing that the symbol of the operator $\mathcal{I}_{\mathcal{S}}$ is a realizable volume polynomial over any field.

Proof of Theorem 1.1 from Theorem 1.3. The polynomial $\sum_{T \in \binom{[n]}{r}} C_T(\mathcal{S}) y^T$ is the result of taking the multi-affine part (i.e. the terms with square-free monomials) of the polynomial $\mathcal{I}_{\mathcal{S}}(\sum_{B \in \binom{[m]}{r}} x_B)$. The elementary symmetric polynomial $\sum_{B \in \binom{[m]}{r}} x_B$ is a realizable volume polynomial over any field because it is over any infinite field $\Bbbk$, witnessed by the arrangement Schubert variety of a generic r-dimensional subspace of $\Bbbk^m$ [**?**, Example 3.5], and taking field extensions do not change the set of realizable volume polynomials [**?**, Proposition 2.10]. The desired conclusion now follows from Theorem 1.3 and that taking the multi-affine part preserves Lorentzian polynomials and realizable volume polynomials [**?**, Corollary 3.5], [**?**, Corollary 3.3]. □

We call the operator $\mathcal{I}_{\mathcal{S}}$ the *inducing operator* of $\mathcal{S}$ because we view it as the polynomial analogue of the construction in matroid theory called *induced (poly)matroids*, as we now explain. A *polymatroid* P on $[m]$ is a function $\mathrm{rk}_{\mathrm{P}} : 2^{[m]} \to \mathbb{Z}_{\geq 0}$ satisfying

(i) $\mathrm{rk}_{\mathrm{P}}(\emptyset) = 0$,
(ii) if $I \subseteq J \subseteq [m]$ then $\mathrm{rk}_{\mathrm{P}}(I) \leq \mathrm{rk}_{\mathrm{P}}(J)$, and
(iii) for all subsets I and J of $[m]$, one has $\mathrm{rk}_{\mathrm{P}}(I) + \mathrm{rk}_{\mathrm{P}}(J) \geq \mathrm{rk}_{\mathrm{P}}(I \cup J) + \mathrm{rk}_{\mathrm{P}}(I \cap J)$.

We call rk_{P} the *rank function* of P, and define the *base polytope* $Q(\mathrm{P})$ of P by

$$Q(\mathrm{P}) := \Big\{ x \in \mathbb{R}^m : \sum_{i \in I} x_i \leq \mathrm{rk}_{\mathrm{P}}(I) \text{ for all } I \subseteq [m] \quad \text{and} \quad \sum_{i \in [m]} x_i = \mathrm{rk}_{\mathrm{P}}([m]) \Big\}.$$

A *matroid* on $[m]$ is a polymatroid whose rank function further satisfies $\mathrm{rk}_{\mathrm{P}}(I) \leq |I|$ for all $I \subseteq [m]$. We point to [**?**] as a standard reference on matroids and polymatroids. Following [**?**, Chapter 8.2], from a polymatroid P on $[m]$ and a sequence of subsets $\mathcal{S} = (S_1, \ldots, S_n)$ of $[m]$, one can construct a new polymatroid $\mathcal{I}_{\mathcal{S}}(\mathrm{P})$ on $[n]$, called the *induced polymatroid*, by setting its rank function to be $\mathrm{rk}_{\mathcal{I}_{\mathcal{S}}(\mathrm{P})}(I) := \mathrm{rk}_{\mathrm{P}}(\bigcup_{i \in I} S_i)$ for all $I \subseteq [n]$. By setting the rank function to be $I \mapsto \min\{|I|, \mathrm{rk}_{\mathrm{P}}(\bigcup_{i \in I} S_i)\}$ instead, one obtains the *induced matroid*.

The operator $\mathcal{I}_{\mathcal{S}}$ gives rise to induced polymatroids as follows. If $f = \sum_\alpha c_\alpha x^\alpha \in \mathbb{R}[x_1, \ldots, x_m]$ is Lorentzian, then its *support*, i.e. $\mathrm{supp}(f) = \{\alpha : c_\alpha \neq 0\}$, is always the set $Q(\mathrm{P}) \cap \mathbb{Z}^m$ of the lattice points of the base polytope of some polymatroid P [**?**, Theorem 2.25]. We say simply that the support of f is the polymatroid P in this case. The Hall–Rado theorem for polymatroids, recalled here as Proposition 2.3, implies that

$$(*) \qquad \mathrm{supp}(f) = Q(\mathrm{P}) \cap \mathbb{Z}^m \implies \mathrm{supp}(\mathcal{I}_{\mathcal{S}}(f)) = Q(\mathcal{I}_{\mathcal{S}}(\mathrm{P})) \cap \mathbb{Z}^n.$$

The operator $\mathcal{I}_{\mathcal{S}}$ is not the only operator satisfying the property ($*$) however; for example, the change of variables $x_i \mapsto \sum_{j:S_j \ni i} y_j$ also has the property ($*$). We point out how $\mathcal{I}_{\mathcal{S}}$ is the "center point" among these in Proposition 3.1.

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2. The symbol of $\mathcal{I}_{\mathcal{S}}$

We use the "symbol theory" originally developed in [**?**] in the context of stable polynomials to prove Theorem 1.3. For $\kappa \in \mathbb{Z}^m_{\geq 0}$, let $\mathbb{R}[x_1, \dots, x_m]_\kappa$ denote the set of polynomials whose support is contained in the box $[0, \kappa_1] \times \cdots \times [0, \kappa_m]$. For $\alpha \in \mathbb{Z}^m_{\geq 0}$ we write $\alpha \leq \kappa$ if $\alpha_i \leq \kappa_i$ for all $i \in [m]$. The *symbol* of a linear operator $T : \mathbb{R}[x_1, \dots, x_m]_\kappa \to \mathbb{R}[y_1, \dots, y_n]$ is defined as

$$sym_T(y, u) := T\big((x_1 + u_1)^{\kappa_1} \cdots (x_m + u_m)^{\kappa_m}\big) \in \mathbb{R}[y_1, \dots, y_n, u_1, \dots, u_m]$$

where T acts trivially on u, or equivalently, $sym_T(y, u) = \sum_{0 \leq \alpha \leq \kappa} \binom{\kappa}{\alpha} T(x^\alpha) u^{\kappa - \alpha}$ where $\binom{\kappa}{\alpha} := \binom{\kappa_1}{\alpha_1} \cdots \binom{\kappa_m}{\alpha_m}$.

Proposition 2.1. Let T be a linear operator $\mathbb{R}[x_1, \dots, x_m]_\kappa \to \mathbb{R}[y_1, \dots, y_n]$ that preserves homogeneous polynomials.

- [**?**, Theorem 3.2] If the symbol of T is Lorentzian, then T preserves Lorentzian polynomials.
- [**?**, Theorem 1.12] If the symbol of T is a realizable volume polynomial over a field $\Bbbk$, then T preserves realizable volume polynomials over $\Bbbk$.

We now fix an arbitrary $\kappa \in \mathbb{Z}^m_{\geq 0}$ for the rest of the section, and consider $\mathcal{I}_{\mathcal{S}}$ as an operator on $\mathbb{R}[x_1, \dots, x_m]_\kappa$. We compute the symbol of $\mathcal{I}_{\mathcal{S}}$ to be

$$sym_{\mathcal{I}_{\mathcal{S}}}(y, u) = \sum_{0 \leq \alpha \leq \kappa} \binom{\kappa}{\alpha} \mathcal{I}_{\mathcal{S}}(x^\alpha) u^{\kappa - \alpha} = \kappa! \sum_{0 \leq \alpha \leq \kappa} \mathcal{I}_{\mathcal{S}}\left(\frac{x^\alpha}{\alpha!}\right) \frac{u^{\kappa - \alpha}}{(\kappa - \alpha)!} = \sum_{0 \leq \alpha \leq \kappa} \sum_{\beta : \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta} \frac{y^\beta}{\beta!} \frac{u^{\kappa - \alpha}}{(\kappa - \alpha)!}.$$

Note that the symbol is the exponential generating function of its support, which is necessarily a polymatroid if the symbol is to be Lorentzian. Our primary goal concerns the linear realizability of this polymatroid. A polymatroid P on $[m]$ is said to be *linearly realizable* over a field $\Bbbk$ if there exist $\Bbbk$-vector spaces $V_1, \dots, V_m$ and a subspace $L \subseteq \bigoplus_{i \in [m]} V_i$ such that

$$\mathrm{rk}_{\mathrm{P}}(A) = \dim \Big(\text{image of } L \text{ under the projection } \bigoplus_{i \in [m]} V_i \to \bigoplus_{j \in A} V_j\Big) \quad \text{for all } A \subseteq [m].$$

Theorem 2.2. The symbol $sym_{\mathcal{I}_{\mathcal{S}}}$ of $\mathcal{I}_{\mathcal{S}}$ is the exponential generating function of the lattice points of the base polytope a polymatroid on $[m] \sqcup [n]$ that is linearly realizable over any infinite field.

This theorem implies our main result Theorem 1.3 as follows.

Proof of Theorem 1.3. [**?**, Corollary 1.4 & Remark 3.21] states that the exponential generating function of a linearly realizable polymatroid over $\Bbbk$ is a realizable volume polynomial over $\Bbbk$. Theorem 2.2 thus implies that $sym_{\mathcal{I}_{\mathcal{S}}}$ is a realizable volume polynomial over any infinite field, and in fact over

any field because the set of realizable volume polynomials do not change under field extensions [**?**, Proposition 2.10]. Realizable volume polynomials are also Lorentzian [**?**, Theorem 4.6]. Proposition 2.1 now implies the theorem. □

We prepare for the proof of Theorem 2.2 with the following variant of the Hall–Rado theorem for polymatroids.

Proposition 2.3. [**?**, Theorem 2] Let P be a polymatroid on a finite set E, and let $\mathcal{E} = (E_1, \dots, E_\ell)$ be a sequence of subsets of E. Then, for $\delta \in \mathbb{Z}^\ell_{\geq 0}$, we have that $\delta \in Q(\mathcal{I}_\mathcal{E}(\mathrm{P}))$ if and only if there exists $\gamma \in \mathbb{Z}^E_{\geq 0}$ such that the pair $(\gamma, \delta) \in \mathbb{Z}^E_{\geq 0} \times \mathbb{Z}^\ell_{\geq 0}$ admits a $\mathcal{E}$-matching.

Proof of Theorem 2.2. We first identify the polymatroid. For an integer r and a finite set $[N]$, let $\Delta^r_{[N]}$ be the polymatroid on $[N]$ whose rank function is $\mathrm{rk}_\mathrm{P}(A) = r$ for all nonempty $A \subseteq [N]$, or equivalently, its base polytope is $Q(\Delta^r_{[N]}) = \{x \in \mathbb{R}^{[N]}_{\geq 0} : \sum_{i\in[N]} x_i = r\}$. It is linearly realizable over any infinite field $\Bbbk$ by a generic embedding of $\Bbbk^r$ in $(\Bbbk^r)^{\oplus[N]}$. Let E be the set of edges of the bipartite graph $\Gamma_\mathcal{S}$ on $[m] \sqcup [n]$, and for each vertex $v \in [m] \sqcup [n]$, let E_v be the set of edges incident to v. Let P be the polymatroid on E defined as the direct sum $\Delta^{\kappa_1}_{E_{i\in[m]}} \oplus \cdots \oplus \Delta^{\kappa_m}_{E_{m\in[m]}}$. Here we wrote $E_{i\in[m]}$ for E_i to avoid possible confusion with $E_{i\in[n]}$. Lastly, let $\mathcal{E}$ be the sequence $(E_v)_{v\in[m]\sqcup[n]}$, and consider the induced polymatroid $\mathcal{I}_\mathcal{E}(\mathrm{P})$ on $[m] \sqcup [n]$, which is linearly realizable over any infinite field $\Bbbk$ because P is.

We claim that a pair $(\alpha, \beta) \in \mathbb{Z}^m_{\geq 0} \times \mathbb{Z}^n_{\geq 0}$ with $0 \leq \alpha \leq \kappa$ admits a $\mathcal{S}$-matching if and only if $(\kappa - \alpha, \beta) \in Q(\mathcal{I}_\mathcal{E}(\mathrm{P}))$. Since $sym_{\mathcal{I}_\mathcal{S}}$ is the exponential generating function of the set of such $(\kappa - \alpha, \beta)$, the theorem follows. Proposition 2.3 implies that $(\kappa - \alpha, \beta) \in Q(\mathcal{I}_\mathcal{E}(\mathrm{P}))$ if and only if there exists $\gamma \in Q(\mathrm{P}) \subset \mathbb{Z}^E_{\geq 0}$ such that γ admits a $\mathcal{E}$-matching to $(\kappa - \alpha, \beta)$. The edge weighting of $\Gamma_\mathcal{E}$ witnessing such a $\mathcal{E}$-matching can be viewed as a function $\mathbf{w} : E \to \mathbb{Z}^2_{\geq 0}$, $e \mapsto (w_e, w'_e)$ on the edges of $\Gamma_\mathcal{S}$ such that:

- $(w_e + w'_e)_{e\in E} \in Q(\mathrm{P})$,
- for every $i \in [m]$, the sum of w_e over all $e \in E$ incident to i is $\kappa_i - \alpha_i$, and
- for every $j \in [n]$, the sum of w'_e over all $e \in E$ incident to j is β_j.

Since P is the direct sum the $\Delta^{\kappa_i}_{E_{i\in[m]}}$, we have $Q(\mathrm{P}) = \prod_{i\in[m]} Q(\Delta^{\kappa_i}_{E_{i\in[m]}})$, and thus $(w_e + w'_e)_{e\in E} \in Q(\mathrm{P})$ if and only if $\sum_{e\in E_{i\in[m]}}(w_e + w'_e) = \kappa_i$ for every $i \in [m]$. Hence, the last two properties of $\mathbf{w}$ imply that the edge weighting $w' : E \to \mathbb{Z}_{\geq 0}$, $e \mapsto w'_e$ witnesses that (α, β) admits a $\mathcal{S}$-matching. Conversely, if (α, β) admits a $\mathcal{S}$-matching witnessed by an edge weighting $w' : E \to \mathbb{Z}_{\geq 0}$, $e \mapsto w'_e$, then we may extend w' to a function $\mathbf{w} : E \to \mathbb{Z}^2_{\geq 0}$ satisfying the three bulleted properties above by setting $(w_e)_{e\in E}$ to be any nonnegative integer vector satisfying the second bulleted property. □

Remark 2.4. Let E be the edges of the bipartite graph $\Gamma_\mathcal{S}$. Given a function $r : E \to \mathbb{Z}_{\geq 0}$, we say that a pair $(\alpha, \beta) \in \mathbb{Z}^m_{\geq 0} \times \mathbb{Z}^n_{\geq 0}$ admits a *r-restricted $\mathcal{S}$-matching* if it admits a $\mathcal{S}$-matching witnessed by an edge weighting $w : E \to \mathbb{Z}_{\geq 0}$ satisfying $w_e \leq r_e$ for all $e \in E$. For any r and κ, the proof of Theorem 2.2 modifies straightforwardly to show that the symbol of the analogue of the operator $\mathcal{I}_\mathcal{S}$ for r-restricted $\mathcal{S}$-matchings is also a realizable volume polynomial over any field. The only required change is to replace the polymatroid $\Delta^{\kappa_i}_{E_{i\in[m]}}$ by the polymatroid that is linearly realizable by a generic embedding of $\Bbbk^{\kappa_i}$ in $\bigoplus_{e\in E_{i\in[m]}} \Bbbk^{r_e}$.

A polynomial $f(x_1, \ldots, x_n)$ is *stable* if it has no roots in the upper half-plane of $\mathbb{C}^n$. Exponential generating functions of polymatroids are often not stable. In agreement, the operator $\mathcal{I}_{\mathcal{S}}$ often does not preserve stability.

Example 2.5. Let $\mathcal{S} = (123, 123, 12)$ be a sequence of subsets of $[3]$. By noting that every $\beta \in \mathbb{Z}^3_{\geq 0}$ with $\beta_1 + \beta_2 + \beta_3 = 3$ except $(0, 0, 3)$ admits a $\mathcal{S}$-matching to $(1, 1, 1)$, one computes that

$$\mathcal{I}_{\mathcal{S}}(x_1x_2x_3) = \frac{1}{6}((y_1 + y_2 + y_3)^3 - y_3^3).$$

We have a factorization

$$(y_1 + y_2 + y_3)^3 - y_3^3 = (y_1 + y_2)((y_1 + y_2 + y_3)^2 + (y_1 + y_2 + y_3)y_3 + y_3^2) = (y_1 + y_2)y_3^2(v^2 + v + 1)$$

where $v = (y_1 + y_2 + y_3)/y_3$. Note that the roots of $v^2 + v + 1$ lie in the upper half plane of $\mathbb{C}$. Using this, we can construct a solution to $I_{\mathcal{S}}(x_1x_2x_3) = 0$ such as $y_3 = e^{\frac{\pi i}{12}}$ and $y_1 = y_2 = \frac{1}{2}(e^{\frac{3\pi i}{4}} - e^{\frac{\pi i}{12}})$, which witnesses that $\mathcal{I}_{\mathcal{S}}(x_1x_2x_3)$ is not stable.

Remark 2.6. We note the following generalization of Theorem 1.1, which can be deduced from Theorem 1.3 similarly. For a matroid M on $[m]$ of rank r, and a subset $T \subseteq [n]$, let

$$C_T(\mathrm{M}, \mathcal{S}) := |\{B \text{ a basis of } \mathrm{M} : \text{there is a bijection } m : T \to B \text{ with } m(t) \in S_t \text{ for all } t \in T\}|.$$

Then the polynomial $\sum_{T \in \binom{[n]}{r}} C_T(\mathrm{M}, \mathcal{S})y^T$ is Lorentzian, and further a realizable polynomial over a field $\Bbbk$ if the matroid M is realizable over a field of the same characteristic.

3. Inducing operators

Proposition 2.3 implies that the linear operator has the following support property ($*$):

> If the support of a Lorentzian polynomial $f \in \mathbb{R}[x_1, \ldots, x_m]$ is a polymatroid P, then the support of the polynomial $\mathcal{I}_{\mathcal{S}}(f)$ is the induced polymatroid $\mathcal{I}_{\mathcal{S}}(\mathrm{P})$.

There are other linear operators with this property ($*$), such as the following.

Let $A_{\mathcal{S}}$ be the matrix whose columns are $\mathbf{e}_{S_1}, \ldots, \mathbf{e}_{S_n}$, where $\mathbf{e}_S := \sum_{i \in S} \mathbf{e}_i \in \mathbb{R}^m$. The linear operator $A_{\mathcal{S}} : f(x) \mapsto f(A_{\mathcal{S}}y)$, or equivalently the change of variables operator $x_i \mapsto \sum_{S_j \ni i} y_j$, satisfies the support property ($*$). Indeed, it is straightforward to verify that the support of $A_{\mathcal{S}}(x^\alpha)$ and that of $\mathcal{I}_{\mathcal{S}}(x^\alpha)$ coincide for $\alpha \in \mathbb{Z}^m_{\geq 0}$. Combinatorially, the distinction between $\mathcal{I}_{\mathcal{S}}$ and $A_{\mathcal{S}}$ is that for $A_{\mathcal{S}}$, the coefficient of $y^\beta/\beta!$ in $A_{\mathcal{S}}(x^\alpha/\alpha!)$ is given by the number of edge weightings of $\Gamma_{\mathcal{S}}$ witnessing that $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$, where $\mathcal{I}_{\mathcal{S}}$ only detects the existence of such an edge weighting.

More generally, let A be a nonnegative $m \times n$ matrix whose zero-nonzero pattern is the same as that of the matrix whose columns are $\mathbf{e}_{S_1}, \ldots, \mathbf{e}_{S_n}$, where $\mathbf{e}_S := \sum_{i \in S} \mathbf{e}_i \in \mathbb{R}^m$. Then the linear operator $A : f(x) \mapsto f(Ay)$ satisfies the support property ($*$). The symbol of A is Lorentzian, being a product of nonnegative linear forms.

Proposition 3.1. Fix any $\kappa \in \mathbb{Z}^m_{\geq 0}$. Let $\mathbf{I}_{\mathcal{S}}$ be the space of linear operators $T : \mathbb{R}[x]_\kappa \to \mathbb{R}[y]$ with Lorentzian symbol and with property ($*$). Then $\mathbf{I}_{\mathcal{S}}$ deformation retracts to the point $\mathcal{I}_{\mathcal{S}}$.

Proof. The symbol of such an operator $T \in \mathbf{I}_{\mathcal{S}}$ is of the form

$$\sum_{\substack{\alpha,\beta:0\leq\alpha\leq\kappa \\ \text{and } \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta}} c_{\alpha,\beta} \frac{y^\beta}{\beta!} \frac{u^{\kappa-\alpha}}{(\kappa-\alpha)!}$$

for some $c_{\alpha,\beta} > 0$. For any $0 \leq q \leq 1$, define T^q to be the operator whose symbol is

$$\sum_{\substack{\alpha,\beta:0\leq\alpha\leq\kappa \\ \text{and } \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta}} c_{\alpha,\beta}^q \frac{y^\beta}{\beta!} \frac{u^{\kappa-\alpha}}{(\kappa-\alpha)!}.$$

We have $T^0 = \mathcal{I}_{\mathcal{S}}$. For $0 \leq q \leq 1$, taking q-th power of normalized coefficients of a Lorentzian polynomial yields a Lorentzian polynomial [**?**, Proposition 3.25], so $T^q \in \mathbf{I}_{\mathcal{S}}$ for all $0 \leq q \leq 1$. □

Since the symbol of $A_{\mathcal{S}}$ is Lorentzian, the proof of the proposition provides a quick way to verify that the symbol of $\mathcal{I}_{\mathcal{S}}$ is Lorentzian. However, while the symbol of $A_{\mathcal{S}}$ is a realizable volume polynomial (over any field), being a product of nonnegative linear forms, one cannot conclude similarly that $sym_{\mathcal{I}_{\mathcal{S}}}$ is a realizable volume polynomial. In fact, we ask:

Question 3.2. For a polynomial $f = \sum_\alpha c_\alpha \frac{x^\alpha}{\alpha}$, denote $f^q := \sum_\alpha c_\alpha^q \frac{x^\alpha}{\alpha}$. If f^1 is a (realizable) volume polynomial (over a field $\Bbbk$), is f^q a volume polynomial (over $\Bbbk$) for all $0 \leq q \leq 1$?

One may seek a different 1-parameter family of operators in $\mathbf{I}_{\mathcal{S}}$, consisting of operators whose symbols are volume polynomials, that connects $\mathcal{I}_{\mathcal{S}}$ and $A_{\mathcal{S}}$. The following example shows that this cannot be achieved by varying the coefficients of the matrix A.

Example 3.3. Let $\mathcal{S} = (1, 2, 12)$ be a sequence of subsets of $[2]$. We have $\mathcal{I}_{\mathcal{S}}(x_1x_2) = y_1y_2 + y_1y_3 + y_2y_3 + \frac{y_3^2}{2}$, whereas $A(x_1x_2)$ is of the form $(ay_1 + by_3)(cy_2 + dy_3) = acy_1y_2 + ady_1y_3 + bcy_2y_3 + bdy_3^2$. For all values of (a, b, c, d), the latter has the property that the product of the coefficients of y_1y_2 and y_3^2 is equal to the product of the coefficients of y_1y_3 and y_2y_3.

The desired family of operators can be achieved using Theorem 2.2 as follows. For $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$, say $S_i = \left\{j_1^{(i)}, \ldots, j_{N_i}^{(i)}\right\}$, define another sequence $\widetilde{\mathcal{S}}$ of subsets of $[m]$ by

$$\widetilde{\mathcal{S}} = (S_1, \ldots, S_n; j_1^{(1)}, \ldots, j_{N_1}^{(1)}, \ldots, j_1^{(n)}, \ldots, j_{N_n}^{(n)}).$$

Let $N = \sum_{i=1}^n N_i$. By Theorem 2.2, the symbol $sym_{\mathcal{I}_{\widetilde{\mathcal{S}}}}(y_1, \ldots, y_n, \widetilde{y}_1, \ldots, \widetilde{y}_N, u_1, \ldots, u_m)$ is a realizable volume polynomial over any field $\Bbbk$. Let $\boldsymbol{y}'$ be the vector $(y_1, \ldots, y_1, y_2, \ldots, y_2, \ldots, y_n, \ldots, y_n)$ where y_i is repeated N_i many times. For any nonnegative vectors $\boldsymbol{a} \in \mathbb{R}^n_{\geq 0}$ and $\boldsymbol{b} \in \mathbb{R}^N_{\geq 0}$, the nonnegative linear change of variables $sym_{\mathcal{I}_{\widetilde{\mathcal{S}}}}(\boldsymbol{a} \cdot \boldsymbol{y}, \boldsymbol{b} \cdot \boldsymbol{y'}, \boldsymbol{u})$ is a volume polynomial over any field. Let $T_{\boldsymbol{a},\boldsymbol{b}}$ be the linear operator with symbol $sym_{\mathcal{I}_{\widetilde{\mathcal{S}}}}(\boldsymbol{a} \cdot \boldsymbol{y}, \boldsymbol{b} \cdot \boldsymbol{y'}, \boldsymbol{u})$. We find that $T_{\mathbf{1},0} = \mathcal{I}_{\mathcal{S}}$ and $T_{0,\mathbf{1}} = A_{\mathcal{S}}$. Thus, any path $(\mathbf{1}, 0) \rightsquigarrow (0, \mathbf{1}) \subseteq \mathbb{R}^n_{\geq 0} \times \mathbb{R}^N_{\geq 0}$ yields the desired 1-parameter family.

Remark 3.4. In [**?**], the authors study the analogue of the induced polymatroids construction for M-convex functions. There, the key strategy is that every bipartite graph can be considered as a composition of simple bipartite graphs, called elementary splits and elementary aggregations. The same strategy cannot be adopted establish the properties of $\mathcal{I}_{\mathcal{S}}$ because the composition of the operators of the form $\mathcal{I}_{\mathcal{S}}$ does not respect composition of bipartite graphs.